DevOps Engineer Roadmap (Beginner to Job-Ready) – 2026

Introduction

DevOps is a combination of software development (Dev) and IT operations (Ops) that focuses on automating software delivery, managing infrastructure, and improving collaboration between development and operations teams. A DevOps Engineer builds CI/CD pipelines, manages cloud infrastructure, automates deployments, monitors applications, and ensures systems are secure, scalable, and highly available. This roadmap provides a practical learning path for beginners aiming to become industry-ready DevOps Engineers.

Estimated Duration: 8–12 Months (2–4 hours/day)

Phase 1: Linux, Networking & Programming Fundamentals (1–2 Months)

Build a strong foundation in operating systems, networking, and scripting.

Learn

- Linux Command Line
- File System & Permissions
- Bash Scripting
- Python Basics
- Git & GitHub
- Computer Networks (TCP/IP, DNS, HTTP/HTTPS)
- Operating System Fundamentals

Mini Projects

- Linux Automation Scripts
- Log File Analyzer
- Server Health Monitoring Script

Phase 2: Version Control & Software Development Basics (3 Weeks)

Understand how developers build and manage software projects.

Learn

- Git Workflow
- GitHub
- Branching & Merging
- Pull Requests
- Software Development Lifecycle (SDLC)
- Agile & Scrum
- REST APIs (Basics)

Project

- Manage a Team Project Using GitHub

Phase 3: Docker & Containerization (1 Month)

Learn how to package and run applications consistently across environments.

Learn

- Docker Architecture
- Docker Images & Containers
- Dockerfile
- Docker Compose
- Docker Networking
- Docker Volumes
- Multi-stage Builds

Project

- Dockerize a Web Application

Phase 4: CI/CD Pipelines (1 Month)

Automate software building, testing, and deployment.

Learn

- Continuous Integration (CI)
- Continuous Deployment (CD)
- Pipeline Design
- Build Automation
- Deployment Strategies
- Secrets Management

Tools

- GitHub Actions
- Jenkins
- GitLab CI/CD (Basics)

Project

- Build a Complete CI/CD Pipeline

Phase 5: Cloud Computing (1–2 Months)

Learn how modern applications are deployed and managed in the cloud.

Learn

- Cloud Fundamentals
- Virtual Machines
- Storage Services
- Networking
- IAM
- Load Balancers
- Auto Scaling

Cloud Platforms

- AWS
- Microsoft Azure
- Google Cloud Platform

Project

- Deploy a Web Application on AWS

Phase 6: Kubernetes & Container Orchestration (1–2 Months)

Manage and scale containerized applications using Kubernetes.

Learn

- Kubernetes Architecture
- Pods
- Deployments
- Services
- ConfigMaps
- Secrets
- Ingress
- Persistent Volumes
- Helm (Basics)

Project

- Deploy a Multi-Container Application on Kubernetes

Phase 7: Infrastructure as Code & Configuration Management (3 Weeks)

Automate infrastructure provisioning and server configuration.

Learn

- Infrastructure as Code (IaC)
- Terraform
- Ansible
- OpenTofu (Introduction)
- Infrastructure Automation

Project

- Provision Cloud Infrastructure Using Terraform

Phase 8: Monitoring, Logging & Security (1 Month)

Learn how to monitor applications and secure production systems.

Learn

- Monitoring
- Logging
- Alerting
- Incident Response
- DevSecOps Basics
- Secrets Management

Tools

- Prometheus
- Grafana
- ELK Stack
- Loki
- HashiCorp Vault (Basics)

Project

- Build a Monitoring Dashboard for a Kubernetes Cluster

Phase 9: Build Your Portfolio

Develop practical projects that demonstrate real-world DevOps skills.

Recommended Portfolio Projects

- Dockerized Multi-Service Application
- CI/CD Pipeline with GitHub Actions
- Kubernetes Deployment Project
- Infrastructure as Code with Terraform
- AWS Cloud Deployment
- Monitoring & Logging Dashboard
- GitOps Deployment using Argo CD (Optional)

Deploy your projects, document your architecture, and maintain an active GitHub portfolio.

Phase 10: Interview Preparation

Prepare for technical interviews by strengthening your infrastructure, automation, and cloud knowledge.

Focus Areas

- Linux

- Networking
- Docker
- Kubernetes
- Git
- GitHub Actions
- Jenkins
- CI/CD
- Terraform
- AWS/Azure/GCP
- Monitoring & Logging
- Bash & Python
- System Design Basics
- Behavioral Questions

Practice troubleshooting scenarios, explain your deployment pipelines confidently, and be prepared to discuss infrastructure architecture and automation strategies.

Final Advice

Start by mastering Linux, Git, and networking before moving to Docker, Kubernetes, and cloud platforms. Focus on automation wherever possible and build end-to-end DevOps projects that demonstrate deployment, monitoring, and infrastructure management. Employers value hands-on experience with CI/CD, containerization, Infrastructure as Code, and cloud services. A well-documented GitHub portfolio featuring production-ready DevOps projects will greatly improve your chances of securing a DevOps Engineer role in 2026.